EXP-5 COMPARATIVE ANALYSIS OF VGG, RESNET AND GOOGLENET ON IMAGE CLASSIFICATION

AIM:

To implement and compare the performance of VGG, ResNet, and GoogleNet architecture for image classification using a deep learning model

SOFTWARE REQUIREMENTS:

- Python 3.x
- Pytorch
- Torch vision
- Matplotlib
- CIFAR-10 Dataset

ALGORITHM:

- Import the required libraries
- Load and preprocess the CIFAR-10 dataset
- Load pretrained VGG16, ResNet 18, and GoogleNet models
- Modify the final classification layers for 10 classes
- Train each model using the Adam optimizer
- Calculate the training and validation accuracy
- plot the accuracy comparison
- Compare the performance of all three models.

PROGRAM:

```
import torch
import torch.nn as nn
import torch.vision
import torchvision.transforms as transforms
from torchvision import models
from torch.utils.data import Dataloader, Subset
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
transform = transforms.Compose([transforms.Resize((224,224)),
        transforms.ToTensor(), transforms.Normalize((0.5)*3,(0.5)*3)
])

dataset = torchvision.datasets.CIFAR10(
        root="./data", train=True,
        download=True, transform=transform)

dataset = Subset(dataset, range(500))
loader = DataLoader(dataset, batch_size=16, shuffle=True)

def get_model(name):
    if name == "vgg":
        model = models.vgg16(weights="DEFAULT")
        model.classifier[6] = nn.Linear(
            model.classified[6].in_feature, 10)
    return model.to(device)

def train_model(model, name):
    optimizer = optim.Adam(model.parameters(), lr=0.0001)
    criterion = nn.CrossEntropyLoss()

    model.train()

    for epoch in range(2):
        correct, total = 0, 0
        for images, labels in loader:
            images, labe = images.to(device), labels.to(device)

            optimizer.zero_grad()
            outputs = model(images)

            if isinstance(outputs, tuple):
                output, labels)
                output = outputs[0]


            accuracy = 100 * correct / total
            print(name, "Epoch:", epoch+1,
                "Accuracy:", round(accuracy,2), "%")
```

Expected Output

Training VGG

VGG epoch: 1 Accuracy: ...

VGG epoch: 2 Accuracy: ...

Training RESNET

RESNET Epoch: 1 Accuracy: ...

RESNET Epoch: 2 Accuracy: ...

Training GOOGLENET

GOOGLENET Epoch: 1 Accuracy: ...

GOOGLENET Epoch: 2 Accuracy: ...

```
for name in ["vgg", "resent", "googlenet"]:
    print("\nTraining", name.upper())
    model = get_model(name)
    train_model(model, name.upper())
```

RESULT:

VGG16, ResNet 18 and GoogleNet were implemented for image classification. Their training performance was compared to understand the effect of network depth residual connections, and inception modules.